Groups and Lie Algebras
Lecture Notes

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Chapter 1

Symmetries

Example.

In physics symmetries are often first introduced in the context of classical mechanics in order to simplify computations.

Definition 1.0.1: Symmetry

A **symmetry** is an invertible map

$$s : \mathbb{R}^n \to \mathbb{R}^n$$

such that, whenever $x(t)$ is a solution of the equations of motion, also

$$s \circ x(t)$$

is a solution.

Example.

Consider a free particle, so that $\vec{F} = 0$. Then the equations of motion have solutions

$$x(t) = x_0 + tv_0, \qquad x_0, v_0 \in \mathbb{R}^n.$$

For any $T \in \mathbb{R}^n$, the translation

$$s_T : \mathbb{R}^n \to \mathbb{R}^n, \qquad x \mapsto x + T$$

is a symmetry.

Example.

In quantum mechanics, a symmetry is typically a unitary operator

$$S : \mathcal{H} \to \mathcal{H}$$

on the Hilbert space $\mathcal{H}$ such that

$$[S, H] = 0,$$

where $H : \mathcal{H} \to \mathcal{H}$ is the Hamiltonian.

For

$$\mathcal{H} = L^2(\mathbb{R}), \qquad H = -\frac{1}{2m}\partial_x^2,$$

the translation operator

$$S_T : L^2(\mathbb{R}) \to L^2(\mathbb{R}), \qquad (S_T f)(x) = f(x+T),$$

and the reflection operator

$$S_R : L^2(\mathbb{R}) \to L^2(\mathbb{R}), \qquad (S_R f)(x) = f(-x),$$

are symmetries.

Abstract structure	**Concrete realization**
group G	representation ρ_V
$g \in G$	$\rho(g) : V \to V$
$e \in G$ (“do nothing”)	$\rho(e) = \mathrm{id}_V$
$h \circ g \in G$ (symmetry composition)	$\rho(h \circ g) = \rho(h)\,\rho(g)$
$\exists\, g^{-1}$ s.t. $g^{-1} \circ g = g \circ g^{-1} = e$	$\rho(g^{-1})\,\rho(g) = \rho(g)\,\rho(g^{-1}) = \mathrm{id}_V$

Definition 1.0.2: Group

A **group** is a set G together with a map

$$\circ : G \times G \to G, \qquad (g, h) \mapsto g \circ h,$$

such that:

1. $\circ$ is associative, that is,

$$(g \circ h) \circ k = g \circ (h \circ k) \qquad \forall g, h, k \in G.$$

2. There exists an element $e \in G$ such that

$$e \circ g = g \circ e = g \qquad \forall g \in G.$$

3. For every $g \in G$, there exists an element $g^{-1} \in G$ such that

$$g \circ g^{-1} = g^{-1} \circ g = e.$$

Corollary 1.0.3

In a group, the unit element and inverses are unique.

Proof:

Suppose e and e' are both unit elements. Then

$$e = e \circ e' = e'.$$

Hence the unit is unique.

Now let $g \in G$, and suppose h and h' are both inverses of g. Then

$$h = h \circ e = h \circ (g \circ h') = (h \circ g) \circ h' = e \circ h' = h'.$$

Hence the inverse of any element is unique.

Definition 1.0.4: Representation

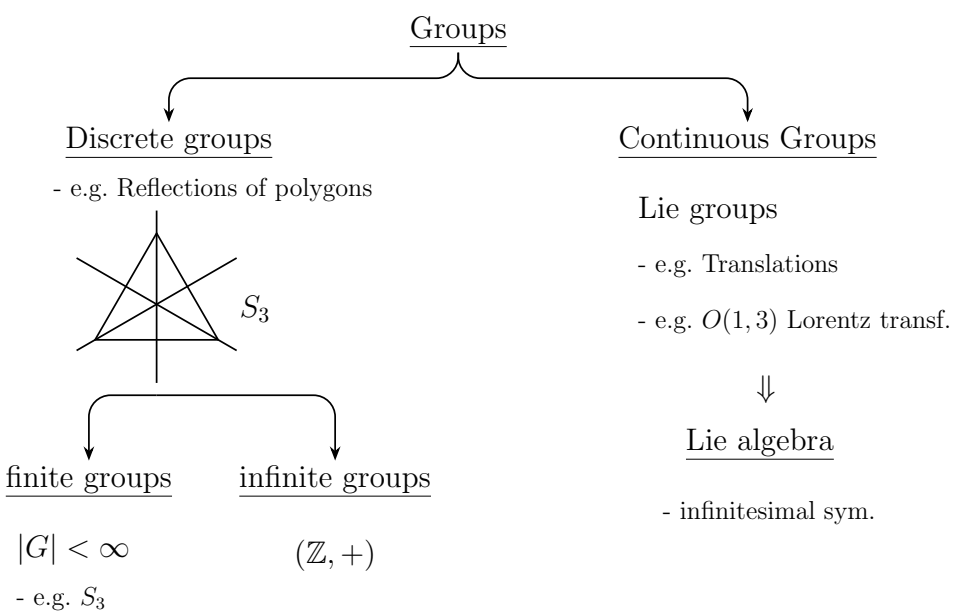
Let $(G, \circ)$ be a group. Usually we simply write G and leave the map $\circ$ implicit. A **representation** of G is a vector space V together with a map

$$\rho : G \to \mathrm{End}(V)$$

such that:

1. $\rho(e) = \mathrm{id}_V$.
2. $\rho(g)\rho(h) = \rho(g \circ h) \qquad \forall g, h \in G.$

We will study these two definitions in this course. The basic structure of the course is as follows:



Chapter 2

Basic group theory

2.1 Structure preserving maps

Definition 2.1.1: Group homomorphism

Let G and H be groups. A **group homomorphism** is a map

$$f : G \to H$$

such that

$$f(g_1 \circ_G g_2) = f(g_1) \circ_H f(g_2) \qquad \forall g_1, g_2 \in G.$$

Corollary 2.1.2

If $f : G \to H$ is a group homomorphism, then

$$f(e_G) = e_H.$$

Corollary 2.1.3

If $f : G \to H$ and $\tilde{f} : H \to K$ are group homomorphisms, then

$$\tilde{f} \circ f : G \to K$$

is again a group homomorphism.

Corollary 2.1.4

If $f : G \to H$ is a bijective group homomorphism, then

$$f^{-1} : H \to G$$

is also a group homomorphism.

Corollary 2.1.5

Every group homomorphism preserves inverses, i.e. if $f : G \to H$ is a group homomorphism, then

$$f(g^{-1}) = f(g)^{-1} \qquad \forall g \in G.$$

Proof:

Since $gg^{-1} = e_G$, we get

$$f(g)f(g^{-1}) = f(gg^{-1}) = f(e_G) = e_H.$$

Similarly,

$$f(g^{-1})f(g) = f(g^{-1}g) = f(e_G) = e_H.$$

Hence $f(g^{-1})$ is the inverse of $f(g)$, so $f(g^{-1}) = f(g)^{-1}$.

Definition 2.1.6: Group isomorphism

An invertible group homomorphism is called a **group isomorphism**. If there exists a group isomorphism

$$f : G \to H$$

then we say that G and H are **isomorphic**.

Example.

Examples of discrete groups:

- $(\mathbb{Z}, +)$.
- $(\mathbb{R}, +)$ and $(\mathbb{C}, +)$.
- $(\mathbb{R} \setminus \{0\}, \cdot)$ and $(\mathbb{C} \setminus \{0\}, \cdot)$.
- Let $n \in \mathbb{N}_{>0}$. Then $\mathbb{Z}/n\mathbb{Z}$ is the quotient of $\mathbb{Z}$ by the equivalence relation

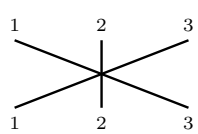
  $$a \sim b \qquad \Longleftrightarrow \qquad a = b + kn \quad \text{for some } k \in \mathbb{Z}.$$

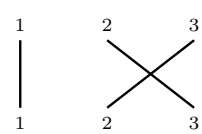
  The equivalence class of a is $[a] = \{b \in \mathbb{Z} \mid b \sim a\}$, and $\mathbb{Z}/n\mathbb{Z}$ becomes a group with operation

  $$[a] \cdot [b] = [a + b].$$

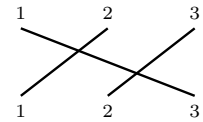
- The permutation group S_n, for $n \in \mathbb{N}_{>0}$, is the group of all permutations of $\{1, 2, \ldots, n\}$. With composition as multiplication, S_n forms a group, and $|S_n| = n!$.

  For example, in S_3 let

$\sigma_1 =$ 

, $\sigma_2 =$ 

Then

$\sigma_2 \circ \sigma_1 =$ 

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Example.

Examples of continuous groups:

- The general linear group

$$\mathrm{GL}(n, \mathbb{R}) = \{M : \mathbb{R}^n \to \mathbb{R}^n \mid M \text{ linear}, \ \det M \neq 0\}.$$

- The orthogonal group

$$\mathrm{O}(n) = \{M : \mathbb{R}^n \to \mathbb{R}^n \mid M \text{ linear}, \ M^T M = I\}.$$

  Equivalently, $M \in \mathrm{O}(n)$ iff $\det M = \pm 1$.

- More generally, let $T_{ij} \in T\mathbb{R}^{n*} \otimes T\mathbb{R}^{n*}$ be a tensor. Its symmetry group is

$$\mathrm{Sym}(T_{ij}) = \{M^i{}_j \in \mathrm{GL}(n, \mathbb{R}) \mid T_{ij} = T_{ab} M^a{}_i M^b{}_j\}.$$

  If $\langle v, w \rangle = T_{ij} v^i w^j$, this is equivalently the group preserving the bilinear form defined by T.

  For

$$T = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix},$$

  one gets $\mathrm{Sym}(T) = \mathrm{O}(1, 3)$. For the standard symplectic form

$$T = \begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix},$$

  one gets the symplectic group.

- An **affine transformation** is a map $\mathbb{R}^n \to \mathbb{R}^n$ of the form

$$x \mapsto Mx + T, \qquad M \in \mathrm{GL}(n, \mathbb{R}), \ T \in \mathbb{R}^n.$$

  We denote the group of affine transformations by $\mathrm{Aff}(\mathbb{R}^n)$.

- If $B \subseteq \mathbb{R}^n$, define

$$\mathrm{Sym}(B) = \{A \in \mathrm{Aff}(\mathbb{R}^n) \mid A(B) = B\}.$$

  For an equilateral triangle Δ, one has $\mathrm{Sym}(\Delta) \cong S_3$.

Definition 2.1.7: Subgroup

Let G be a group. A **subgroup** $H \leq G$ is a subset $H \subseteq G$ such that H is itself a group:

1. $h \circ_G h' \in H$ for all $h, h' \in H$,
2. if $h \in H$, then $h^{-1} \in H$.

Example.

Examples:

- $\mathrm{SO}(n) \leq \mathrm{O}(n) \leq \mathrm{GL}(n)$.
- $S_2 \leq S_3 \leq S_4$.

This notation is slightly sloppy: strictly speaking, one should specify an injective group homomorphism $H \hookrightarrow G$. For example,

$$H = \{\sigma \in S_4 \mid \sigma(4) = 4\}$$

is isomorphic to S_3.

Proposition 2.1.8

Let $f : G \to H$ be a group homomorphism. Then the kernel of f,

$$\ker f := \{g \in G \mid f(g) = e_H\} \leq G,$$

is a subgroup. Likewise, the image of f,

$$\operatorname{im} f := \{h \in H \mid \exists g \in G,\ f(g) = h\} \leq H,$$

is a subgroup.

Proof:

Let $g_1, g_2 \in \ker f$. Then

$$f(g_1 \circ g_2) = f(g_1)f(g_2) = e_H,$$

so $g_1 \circ g_2 \in \ker f$. Also,

$$f(g^{-1}) = f(g)^{-1} = e_H$$

for every $g \in \ker f$, hence $g^{-1} \in \ker f$. Therefore $\ker f$ is a subgroup.

Now let $h_1, h_2 \in \operatorname{im} f$. Then there exist $g_1, g_2 \in G$ such that $f(g_1) = h_1$ and $f(g_2) = h_2$. Hence

$$h_1 h_2 = f(g_1)f(g_2) = f(g_1 \circ g_2) \in \operatorname{im} f.$$

Moreover, if $h = f(g) \in \operatorname{im} f$, then

$$h^{-1} = f(g)^{-1} = f(g^{-1}) \in \operatorname{im} f.$$

Thus $\operatorname{im} f$ is a subgroup.

Example.

Let $\Lambda \subseteq \mathbb{R}^n$ be a lattice. Then there is a group homomorphism

$$\operatorname{Sym}(\Lambda) \to \mathrm{O}(n), \qquad f(x) = Ax + b \mapsto A.$$

The kernel is the group of translational symmetries of Λ, and the image is called the **point group**.

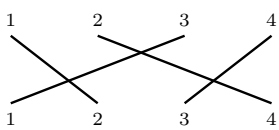
Example.

There is a group homomorphism

$$\operatorname{sign} : S_n \to \{\pm 1\}.$$

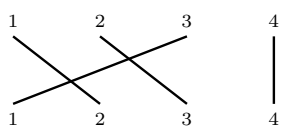
It records whether a permutation is even or odd: even permutations are sent to 1, odd permutations to -1.

For example,

$\sigma =$ 

$\Rightarrow \qquad \operatorname{sign}(\sigma) = -1,$

while

$\tilde{\sigma} =$ 

$\Rightarrow \qquad \operatorname{sign}(\tilde{\sigma}) = 1.$

Definition 2.1.9: Alternating group

The **alternating group** A_n is the kernel of the sign homomorphism:

$$A_n := \ker(\operatorname{sign}) \leq S_n.$$

Equivalently, A_n is the subgroup of even permutations.

Example.

For $n = 3$,

$$A_3 = \left\{ \begin{tikzcd} 1 \arrow[d, dash] & 2 \arrow[d, dash] & 3 \arrow[d, dash] \\ 1 & 2 & 3 \end{tikzcd}, \begin{tikzcd} 1 \arrow[dr, dash] & 2 \arrow[dr, dash] & 3 \arrow[dl, dash] \\ 1 & 2 & 3 \end{tikzcd}, \begin{tikzcd} 1 \arrow[dr, dash] & 2 \arrow[dl, dash] & 3 \arrow[dr, dash] \\ 1 & 2 & 3 \end{tikzcd} \right\}.$$

Hence A_3 has three elements and is isomorphic to $\mathbb{Z}/3\mathbb{Z}$.

2.2 Conjugacy Classes

Definition 2.2.1: Conjugation

Let G be a group and $g \in G$. Define the group homomorphism

$$\mathrm{Ad}_g : G \to G, \qquad h \mapsto ghg^{-1}.$$

This is called **conjugation by** g.
Indeed,

$$\mathrm{Ad}_g(e) = geg^{-1} = e,$$

and

$$\mathrm{Ad}_g(hh') = ghh'g^{-1} = ghg^{-1}gh'g^{-1} = \mathrm{Ad}_g(h)\,\mathrm{Ad}_g(h').$$

Definition 2.2.2: Conjugate Elements

Let $h, h' \in G$. We say that h and h' are **conjugate** if there exists $g \in G$ such that

$$h = gh'g^{-1} = \mathrm{Ad}_g(h').$$

Proposition 2.2.3

Being conjugate is an equivalence relation on G.

Proof:

- Reflexivity: every element is conjugate to itself, since $h = ehe^{-1}$.
- Symmetry: if $h \sim h'$, so $h = gh'g^{-1}$, then $h' = g^{-1}hg$, hence $h' \sim h$.
- Transitivity: if $h_1 \sim h_2$ and $h_2 \sim h_3$, say $h_1 = g_1h_2g_1^{-1}$ and $h_2 = g_2h_3g_2^{-1}$, then

  $$h_1 = g_1g_2h_3(g_1g_2)^{-1},$$

  so $h_1 \sim h_3$.

Definition 2.2.4: Conjugacy Classes

The equivalence classes of the conjugation relation are called **conjugacy classes**. The set of all conjugacy classes is denoted by $\mathrm{conj}(G)$.

Example.

The conjugacy classes of S_3 are

$$\mathrm{conj}(S_3) = \left\{ \{e\}, \left\{ \begin{tikzcd} 1 \arrow[dr] & 2 \arrow[dl] \\ 1 & 2 \end{tikzcd}, \begin{tikzcd} 3 \arrow[d, dash] \\ 3 \end{tikzcd}, \begin{tikzcd} 1 \arrow[dr] & 2 \arrow[d, dash] & 3 \arrow[dl] \\ 1 & 2 & 3 \end{tikzcd}, \begin{tikzcd} 1 \arrow[d, dash] \\ 1 \end{tikzcd}, \begin{tikzcd} 2 \arrow[dr] & 3 \arrow[dl] \\ 2 & 3 \end{tikzcd} \right\}, \left\{ \begin{tikzcd} 1 \arrow[dr] & 2 \arrow[dr] & 3 \arrow[dl] \\ 1 & 2 & 3 \end{tikzcd}, \begin{tikzcd} 1 \arrow[dr] & 2 \arrow[dl] & 3 \arrow[dr] \\ 1 & 2 & 3 \end{tikzcd} \right\} \right\}.$$

Definition 2.2.5: Abelian Group

A group G is called **abelian** if

$$hg = gh \qquad \forall h, g \in G.$$

Corollary 2.2.6

$$\mathrm{conj}(G) = G \qquad \Longleftrightarrow \qquad G \text{ is abelian.}$$

Definition 2.2.7: Cycle

For $1 \leq k \leq n$, let $a_1, \ldots, a_k \in \{1, 2, \ldots, n\}$ be pairwise distinct. A **cycle** $(a_1, \ldots, a_k)$ is the permutation

$$a_1 \mapsto a_2, \quad a_2 \mapsto a_3, \quad \ldots, \quad a_k \mapsto a_1,$$

which acts as the identity on all other elements. We call k the **length** of the cycle.

Example.

To describe $\mathrm{conj}(S_n)$, it is useful to use cycle type.

For $\sigma \in S_n$, the **cycle type** of σ is the list of lengths of the cycles in a disjoint cycle decomposition of σ, written in decreasing order. Thus the cycle type is a partition

$$\lambda_1 \geq \cdots \geq \lambda_k > 0, \qquad \sum_{i=1}^{k} \lambda_i = n.$$

For example, $(1\,3\,2)(4\,5) \in S_5$ has cycle type $(3, 2)$.

Theorem 2.2.8: Conjugacy In S_n

Two permutations $\sigma, \sigma' \in S_n$ are conjugate if and only if they have the same cycle type.

Proof:

Let $\tau, \sigma \in S_n$. Then

$$(\tau\sigma\tau^{-1})(\tau(i)) = \tau(\sigma(i)).$$

So conjugation by τ simply applies τ to the entries in each cycle. In particular,

$$\tau(a_1 \ldots a_r)\tau^{-1} = (\tau(a_1) \ldots \tau(a_r)).$$

Hence conjugate permutations have the same cycle type.

Conversely, if σ and σ' have the same cycle type, then their disjoint cycles can be matched cycle by cycle. Choosing τ so that it sends the entries of each cycle of σ to the corresponding entries of the matching cycle of σ', we get

$$\tau\sigma\tau^{-1} = \sigma'.$$

Example.

The conjugacy classes of S_4 are in bijection with the partitions of 4:

$$(1, 1, 1, 1), \qquad (2, 1, 1), \qquad (3, 1), \qquad (2, 2), \qquad (4).$$

Representatives are:

- $(1, 1, 1, 1)$: e,
- $(2, 1, 1)$: a transposition, e.g. $(1\,2)$,
- $(3, 1)$: a 3-cycle, e.g. $(1\,2\,3)$,
- $(2, 2)$: a product of two disjoint transpositions, e.g. $(1\,2)(3\,4)$,
- (4): a 4-cycle, e.g. $(1\,2\,3\,4)$.

2.3 Normal Subgroups

Definition 2.3.1: Normal Subgroup

A subgroup $N \leq G$ is called **normal** if

$$gng^{-1} \in N \qquad \forall n \in N,\ \forall g \in G.$$

Example.

In S_3, the subgroup

$$\overline{N} = \{e, (1\,2)\}$$

is not normal, since

$$(1\,3)(1\,2)(1\,3)^{-1} = (2\,3) \notin \overline{N}.$$

On the other hand,

$$N = \{e, (1\,2\,3), (1\,3\,2)\}$$

is a normal subgroup of S_3.

Theorem 2.3.2: Quotient Group

Let $N \leq G$ be a normal subgroup. Let G/N be the set of equivalence classes for the relation $g \sim gn$ for $n \in N$. Then G/N is a group with multiplication

$$[g] \cdot [h] = [gh].$$

Proof:

We first check that the multiplication is well defined. If $g' = gn_1$ and $h' = hn_2$ with $n_1, n_2 \in N$, then

$$g'h' = gn_1hn_2 = gh\,(h^{-1}n_1h)\,n_2.$$

Since N is normal, $h^{-1}n_1h \in N$, hence $(h^{-1}n_1h)n_2 \in N$. Therefore $g'h' \sim gh$, so $[g'][h'] = [gh]$.

The group axioms are inherited from G: associativity follows from that in G, the neutral element is $[e]$, and the inverse of $[g]$ is $[g^{-1}]$.

The classes $[g] = gN \subseteq G$ are also called **cosets**.

Example.

Let $\Lambda \subseteq \mathbb{R}^n$ be a lattice. Then the group of translational symmetries of Λ is normal, and

$$\mathrm{Sym}(\Lambda)/\,\mathrm{Trans}(\Lambda)$$

is the point group.

Another important example is $\mathbb{Z}^2 \leq \mathbb{R}^2$. Then $\mathbb{Z}^2$ is a normal subgroup of $(\mathbb{R}^2, +)$,

and the quotient

$$T^2 = \mathbb{R}^2/\mathbb{Z}^2$$

is the 2-torus.